

Assignment 8 — Write-up

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Part A — Choosing Regularization by Validation

Throughout, $\ell(w, z) \in [0, 1]$ is convex and G -Lipschitz in w w.r.t. $\|\cdot\|$, $\Psi \geq 0$ is α -strongly convex w.r.t. $\|\cdot\|$, the training sample T (size n_T) and validation sample V (size n_V) are independent i.i.d. draws, $L_S(w) = \frac{1}{|S|} \sum_{z \in S} \ell(w, z)$, and $L_{\mathcal{D}}(w) = \mathbb{E}_z \ell(w, z)$.

A.1 Validation selection for a finite random candidate set (10 pts)

Claim. Conditioning on predictors $h_1, \dots, h_K$ independent of V , with $\hat{k} \in \arg \min_{k \in [K]} L_V(h_k)$,

$$\mathbb{E}_V L_{\mathcal{D}}(h_{\hat{k}}) \leq \min_{k \in [K]} L_{\mathcal{D}}(h_k) + 2\sqrt{\frac{\log(2K)}{2n_V}}.$$

Proof. Condition on the h_k 's; they are fixed functions independent of V . For each k ,

$$L_V(h_k) = \frac{1}{n_V} \sum_{z \in V} \ell(h_k, z)$$

is an average of n_V i.i.d. $[0, 1]$ -valued terms with mean $L_{\mathcal{D}}(h_k)$.

Step 1: a uniform deviation bound in expectation. Define

$$\Delta := \max_{k \in [K]} |L_V(h_k) - L_{\mathcal{D}}(h_k)|.$$

By Hoeffding's lemma, each centered term $\ell(h_k, z) - L_{\mathcal{D}}(h_k) \in [-1, 1]$ -ish (a shifted $[0, 1]$ variable) has MGF bounded by $e^{s^2/8}$. Independence across $z \in V$ gives, for the centered average $X_k := L_V(h_k) - L_{\mathcal{D}}(h_k)$,

$$\mathbb{E} e^{sX_k} = \prod_{z \in V} \mathbb{E} e^{(s/n_V)(\ell(h_k, z) - L_{\mathcal{D}}(h_k))} \leq \left(e^{(s/n_V)^2/8}\right)^{n_V} = e^{s^2/(8n_V)},$$

so each X_k (and each $-X_k$) is σ -subgaussian with $\sigma^2 = \frac{1}{4n_V}$. The $2K$ variables $\{\pm X_k\}$ are subgaussian, so by the standard maximal inequality: for any $s > 0$,

$$e^{s\mathbb{E}\Delta} \leq \mathbb{E} e^{s\Delta} = \mathbb{E} \max_{k, \pm} e^{\pm sX_k} \leq \sum_{k, \pm} \mathbb{E} e^{\pm sX_k} \leq 2K e^{s^2\sigma^2/2},$$

hence $\mathbb{E}\Delta \leq \frac{\log(2K)}{s} + \frac{s\sigma^2}{2}$. Optimizing at $s = \sqrt{2\log(2K)}/\sigma$ gives

$$\mathbb{E}_V \Delta \leq \sigma \sqrt{2\log(2K)} = \sqrt{\frac{1}{4n_V}} \sqrt{2\log(2K)} = \sqrt{\frac{\log(2K)}{2n_V}}.$$

Step 2: the selection argument. Let $k^* \in \arg \min_k L_{\mathcal{D}}(h_k)$. Then

$$L_{\mathcal{D}}(h_{\hat{k}}) \stackrel{(i)}{\leq} L_V(h_{\hat{k}}) + \Delta \stackrel{(ii)}{\leq} L_V(h_{k^*}) + \Delta \stackrel{(iii)}{\leq} L_{\mathcal{D}}(h_{k^*}) + 2\Delta,$$

where (i) and (iii) use $|L_V(h_k) - L_{\mathcal{D}}(h_k)| \leq \Delta$, and (ii) uses that $\hat{k}$ minimizes L_V . Taking $\mathbb{E}_V$ and Step 1:

$$\mathbb{E}_V L_{\mathcal{D}}(h_{\hat{k}}) \leq L_{\mathcal{D}}(h_{k^*}) + 2\mathbb{E}_V \Delta \leq \min_{k \in [K]} L_{\mathcal{D}}(h_k) + 2\sqrt{\frac{\log(2K)}{2n_V}}. \quad \square$$

Key insight. Because the h_k are independent of V , the validation loss is an *unbiased* risk estimate for each; the only cost of picking the best is the $\sqrt{\log K}$ entropy of a union bound over K fixed candidates, not the internal complexity of any h_k .

A.2 Oracle inequality over a grid (10 pts)

Claim.

$$\mathbb{E} L_{\mathcal{D}}(h_{\hat{\lambda}}) \leq \min_{\lambda \in \Lambda} \inf_{u \in \mathcal{W}} \left\{ L_{\mathcal{D}}(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T} \right\} + 2\sqrt{\frac{\log(2K)}{2n_V}}.$$

Proof. Apply A.1 conditionally on T . Given T , the K predictors $h_{\lambda} = A_{\lambda}(T)$ ($\lambda \in \Lambda$) are deterministic functions of T , hence independent of V . A.1 (with the candidate set $\{h_{\lambda}\}_{\lambda \in \Lambda}$, K of them) gives

$$\mathbb{E}_V [L_{\mathcal{D}}(h_{\hat{\lambda}}) \mid T] \leq \min_{\lambda \in \Lambda} L_{\mathcal{D}}(h_{\lambda}) + 2\sqrt{\frac{\log(2K)}{2n_V}}.$$

Take $\mathbb{E}_T$ and use $\mathbb{E}_T \min_{\lambda}(\cdot) \leq \min_{\lambda} \mathbb{E}_T(\cdot)$:

$$\mathbb{E} L_{\mathcal{D}}(h_{\hat{\lambda}}) \leq \min_{\lambda \in \Lambda} \mathbb{E}_T L_{\mathcal{D}}(A_{\lambda}(T)) + 2\sqrt{\frac{\log(2K)}{2n_V}}. \quad (*)$$

Week-8 exact-RERM bound. Fix λ and write $w_S = A_{\lambda}(S)$, the exact minimizer of $F_S(w) = L_S(w) + \lambda \Psi(w)$ (which is $\lambda\alpha$ -strongly convex). The movement bound $\|w_S - w_{S'}\| \leq \frac{2G}{\lambda \alpha n_T}$ (one-example change) plus G -Lipschitz ℓ give uniform stability $\beta = \frac{2G^2}{\lambda \alpha n_T}$, so $\mathbb{E}_T [L_{\mathcal{D}}(w_T) - L_T(w_T)] \leq \beta$. Also, for any fixed u ,

$$\mathbb{E}_T L_T(w_T) \leq \mathbb{E}_T [L_T(w_T) + \lambda \Psi(w_T)] = \mathbb{E}_T F_T(w_T) \leq \mathbb{E}_T F_T(u) = L_{\mathcal{D}}(u) + \lambda \Psi(u),$$

using $\Psi \geq 0$, then that w_T minimizes F_T , then $\mathbb{E}_T L_T(u) = L_{\mathcal{D}}(u)$. Adding,

$$\mathbb{E}_T L_{\mathcal{D}}(A_{\lambda}(T)) \leq L_{\mathcal{D}}(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T} \quad \text{for every } u \in \mathcal{W}.$$

Taking $\inf_u$ then $\min_{\lambda}$ and substituting into $(*)$ proves the claim. $\square$

Why the validation term is $\sqrt{\log K/n_V}$. Selecting $\hat{\lambda}$ is exactly the finite-class selection of A.1 with K candidates that are frozen once T is fixed. The validation set certifies each candidate by an unbiased n_V -sample average; controlling the worst of K such averages costs a union bound, contributing $\sqrt{\log K}$ (entropy) times $1/\sqrt{n_V}$ (concentration). It is independent of how complex each RERM rule is, because validation never re-uses T .

A.3 Adapting to an unknown comparator scale (12 pts)

For a comparator u , set $a = \Psi(u)$ and $\lambda_{\text{opt}}(a) = \sqrt{\frac{2G^2}{\alpha n_T}}$.

Optimizing the two-term expression. Define $g(\lambda) = \lambda a + \frac{2G^2}{\lambda \alpha n_T}$. By AM–GM,

$$g(\lambda) \geq 2\sqrt{\lambda a \cdot \frac{2G^2}{\lambda \alpha n_T}} = 2\sqrt{\frac{2G^2 a}{\alpha n_T}} = 2G\sqrt{\frac{2a}{\alpha n_T}},$$

with equality iff $\lambda a = \frac{2G^2}{\lambda \alpha n_T}$, i.e. $\lambda = \lambda_{\text{opt}}(a)$. Moreover at the optimum both summands equal the same value

$$M := \lambda_{\text{opt}}(a) a = \frac{2G^2}{\lambda_{\text{opt}}(a) \alpha n_T} = G\sqrt{\frac{2a}{\alpha n_T}}.$$

The factor-2 discretization. Since $B_{\min}^2 \leq a = \Psi(u) \leq B_{\max}^2$, the grid property gives a $\lambda \in \Lambda$ with $\frac{1}{2}\lambda_{\text{opt}}(a) \leq \lambda \leq 2\lambda_{\text{opt}}(a)$; write $\lambda = c\lambda_{\text{opt}}(a)$ with $c \in [\frac{1}{2}, 2]$. Then

$$g(\lambda) = c\lambda_{\text{opt}}(a) a + \frac{1}{c} \cdot \frac{2G^2}{\lambda_{\text{opt}}(a) \alpha n_T} = \left(c + \frac{1}{c}\right)M.$$

On $[\frac{1}{2}, 2]$ the convex map $c \mapsto c + \frac{1}{c}$ is maximized at the endpoints, where it equals $\frac{5}{2}$. Hence

$$g(\lambda) \leq \frac{5}{2}M = \frac{5}{2}G\sqrt{\frac{2a}{\alpha n_T}} = \frac{5\sqrt{2}}{2}G\sqrt{\frac{\Psi(u)}{\alpha n_T}}.$$

Conclusion. For this λ and this u , the A.2 bracketed term is

$$L_{\mathcal{D}}(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda \alpha n_T} = L_{\mathcal{D}}(u) + g(\lambda) \leq L_{\mathcal{D}}(u) + \frac{5\sqrt{2}}{2}G\sqrt{\frac{\Psi(u)}{\alpha n_T}}.$$

Since A.2 takes the min over $\lambda \in \Lambda$ and inf over comparators, it is bounded by this particular choice, so for every u with $B_{\min}^2 \leq \Psi(u) \leq B_{\max}^2$,

$$\boxed{\mathbb{E} L_{\mathcal{D}}(h_{\hat{\lambda}}) \leq L_{\mathcal{D}}(u) + \frac{5\sqrt{2}}{2}G\sqrt{\frac{\Psi(u)}{\alpha n_T}} + 2\sqrt{\frac{\log(2K)}{2n_V}}.} \quad \square$$

The constant inflated from the oracle value $2\sqrt{2}$ (achievable only if the grid hit $\lambda_{\text{opt}}(a)$ exactly) to $\frac{5}{2}\sqrt{2}$ — the worst case of the factor-2 mismatch $c + 1/c$.

A.4 A dyadic grid and the price of tuning (8 pts)

Construction. Write $C := G\sqrt{2/(\alpha n_T)}$, so $\lambda_{\text{opt}}(a) = C/\sqrt{a}$. As a ranges over $[B_{\min}^2, B_{\max}^2]$, $\lambda_{\text{opt}}(a)$ ranges continuously over

$$[\lambda_{\min}, \lambda_{\max}] = \left[\frac{C}{B_{\max}}, \frac{C}{B_{\min}}\right].$$

Take the dyadic grid

$$\Lambda = \left\{ \lambda_i = \frac{C}{B_{\max}} \cdot 2^i : i = 0, 1, \dots, K-1 \right\}, \quad K = \left\lceil \log_2 \frac{B_{\max}}{B_{\min}} \right\rceil + 1.$$

Consecutive points differ by a factor 2, and the largest is $\lambda_{K-1} = \frac{C}{B_{\max}} 2^{K-1} \geq \frac{C}{B_{\max}} \cdot \frac{B_{\max}}{B_{\min}} = \lambda_{\max}$, so $[\lambda_{\min}, \lambda_{\max}] \subseteq [\lambda_0, \lambda_{K-1}]$. Any target $t = \lambda_{\text{opt}}(a)$ lies in some $[\lambda_i, \lambda_{i+1}] = [\lambda_i, 2\lambda_i]$, giving $\lambda_i \in [t/2, t] \subseteq [t/2, 2t]$ — exactly the A.3 property. Thus

$$K = \left\lceil \log_2 \frac{B_{\max}}{B_{\min}} \right\rceil + 1 = O\left(\log \frac{B_{\max}}{B_{\min}}\right).$$

Equal split $n_T = n_V = n/2$. Substituting into A.3, with $\sqrt{\Psi(u)/(\alpha n/2)} = \sqrt{2\Psi(u)/(\alpha n)}$:

$$\frac{5\sqrt{2}}{2} G \sqrt{\frac{2\Psi(u)}{\alpha n}} = 5 G \sqrt{\frac{\Psi(u)}{\alpha n}}, \quad 2 \sqrt{\frac{\log(2K)}{2 \cdot n/2}} = 2 \sqrt{\frac{\log(2K)}{n}},$$

so

$$\boxed{\mathbb{E} L_{\mathcal{D}}(h_{\hat{\lambda}}) \leq L_{\mathcal{D}}(u) + \underbrace{5 G \sqrt{\frac{\Psi(u)}{\alpha n}}}_{\text{RERM learning}} + \underbrace{2 \sqrt{\frac{\log(2K)}{n}}}_{\text{validation selection}}.}$$

Reading the terms. - **RERM learning term** $5G\sqrt{\Psi(u)/(\alpha n)} = \Theta(1/\sqrt{n})$: the statistical cost of fitting the regularized rule on T . - **Validation-selection term** $2\sqrt{\log(2K)/n}$: the cost of choosing among the K trained rules on V . - **Price of not knowing $\Psi(u)$** . Since $K = O(\log(B_{\max}/B_{\min}))$, the validation term is $O(\sqrt{\log \log(B_{\max}/B_{\min})/n})$ — only *doubly* logarithmic in the unknown scale range. Additionally, the learning constant inflated from $2\sqrt{2} \approx 2.83$ (known scale) to 5. So adaptivity costs a tiny $\sqrt{\log \log}$ additive term plus a modest constant factor, while retaining the optimal $1/\sqrt{n}$ rate.

Part B — Approximate RERM and Optimization Error

Setup: ℓ convex and G -Lipschitz w.r.t. $\|\cdot\|$, $\Psi \geq 0$ α -strongly convex; $F_S(w) = L_S(w) + \lambda\Psi(w)$, exact minimizer w_S , and the algorithm returns $\tilde{w}_S$ with $F_S(\tilde{w}_S) \leq F_S(w_S) + \eta$. Given: movement bound $\|w_S - w_{S'}\| \leq \frac{2G}{\lambda\alpha n}$.

B.1 Approximate minimizers are close to exact minimizers (7 pts)

Claim. $\|\tilde{w}_S - w_S\| \leq \sqrt{2\eta/(\lambda\alpha)}$ for every S .

Proof. F_S is $\lambda\alpha$ -strongly convex. For μ -strongly convex F with minimizer w^* over the convex set $\mathcal{W}$, there is a subgradient $g \in \partial F(w^*)$ with $\langle g, w - w^* \rangle \geq 0$ for all $w \in \mathcal{W}$ (first-order optimality), so

$$F(w) \geq F(w^*) + \langle g, w - w^* \rangle + \frac{\mu}{2} \|w - w^*\|^2 \geq F(w^*) + \frac{\mu}{2} \|w - w^*\|^2.$$

With $\mu = \lambda\alpha$, $w^* = w_S$, $w = \tilde{w}_S$:

$$F_S(w_S) + \frac{\lambda\alpha}{2} \|\tilde{w}_S - w_S\|^2 \leq F_S(\tilde{w}_S) \leq F_S(w_S) + \eta \Rightarrow \frac{\lambda\alpha}{2} \|\tilde{w}_S - w_S\|^2 \leq \eta,$$

i.e. $\|\tilde{w}_S - w_S\| \leq \sqrt{2\eta/(\lambda\alpha)}$. $\square$

B.2 Stability of approximate RERM (8 pts)

Claim. If S, S' differ in one example, then $|\ell(\tilde{w}_S, z) - \ell(\tilde{w}_{S'}, z)| \leq \frac{2G^2}{\lambda\alpha n} + 2G\sqrt{2\eta/(\lambda\alpha)}$.

Proof. Triangle inequality and B.1 (twice) plus the movement bound:

$$\|\tilde{w}_S - \tilde{w}_{S'}\| \leq \|\tilde{w}_S - w_S\| + \|w_S - w_{S'}\| + \|w_{S'} - \tilde{w}_{S'}\| \leq \frac{2G}{\lambda\alpha n} + 2\sqrt{\frac{2\eta}{\lambda\alpha}}.$$

Applying G -Lipschitzness of ℓ ,

$$|\ell(\tilde{w}_S, z) - \ell(\tilde{w}_{S'}, z)| \leq G\|\tilde{w}_S - \tilde{w}_{S'}\| \leq \frac{2G^2}{\lambda\alpha n} + 2G\sqrt{\frac{2\eta}{\lambda\alpha}}. \quad \square$$

B.3 Learning guarantee with optimization error (8 pts)

Claim. $\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda\alpha n} + G\sqrt{2\eta/(\lambda\alpha)}$ for every $u \in \mathcal{W}$.

Proof. $L_{\mathcal{D}}$ is G -Lipschitz (an average of G -Lipschitz $\ell(\cdot, z)$), so by B.1,

$$L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(w_S) + G\|\tilde{w}_S - w_S\| \leq L_{\mathcal{D}}(w_S) + G\sqrt{\frac{2\eta}{\lambda\alpha}}.$$

Take $\mathbb{E}_S$ and apply the exact-RERM bound from A.2 (valid for the exact minimizer w_S with n samples), $\mathbb{E}_S L_{\mathcal{D}}(w_S) \leq L_{\mathcal{D}}(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda\alpha n}$:

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda\alpha n} + G\sqrt{\frac{2\eta}{\lambda\alpha}}. \quad \square$$

The optimization error enters *additively* and only at the slow $\sqrt{\eta}$ rate, through the true-risk Lipschitz gap to the exact minimizer — the stability result B.2 is what makes the exact-RERM term legitimate.

B.4 How accurate must the optimizer be? (7 pts)

Compete with all u with $\Psi(u) \leq B^2$ using $\lambda = \sqrt{2G^2/(\alpha B^2 n)} = \frac{G\sqrt{2}}{B\sqrt{\alpha n}}$.

The two RERM terms balance: $\lambda B^2 = \frac{2G^2}{\lambda\alpha n} = GB\sqrt{2/(\alpha n)}$, summing to $2\sqrt{2} \frac{GB}{\sqrt{\alpha n}}$. Hence

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(u) + 2\sqrt{2} \frac{GB}{\sqrt{\alpha n}} + G\sqrt{\frac{2\eta}{\lambda\alpha}}.$$

Sufficient condition. $G\sqrt{2\eta/(\lambda\alpha)} \leq \frac{GB}{\sqrt{\alpha n}} \iff \frac{2\eta}{\lambda\alpha} \leq \frac{B^2}{\alpha n} \iff \eta \leq \frac{\lambda B^2}{2n}$. Substituting λ :

$$\boxed{\eta \leq \frac{\lambda B^2}{2n} = \frac{GB}{\sqrt{2\alpha} n^{3/2}} = O\left(\frac{GB}{\sqrt{\alpha} n^{3/2}}\right).}$$

Resulting excess risk. Under this condition,

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) - L_{\mathcal{D}}(u) \leq 2\sqrt{2} \frac{GB}{\sqrt{\alpha n}} + \frac{GB}{\sqrt{\alpha n}} = (2\sqrt{2} + 1) \frac{GB}{\sqrt{\alpha n}} = O\left(\frac{GB}{\sqrt{\alpha n}}\right).$$

So an optimizer accurate to $\eta = O(n^{-3/2})$ leaves the statistical $1/\sqrt{n}$ rate intact: optimization error is essentially free.

B.5 Soft-margin linear classification (5 pts)

Here $\ell(w, (x, y)) = (1 - y\langle w, x \rangle)_+$ with $\|x\|_2 \leq R$, and $\Psi(w) = \frac{1}{2}\|w\|_2^2$, so $\|\cdot\| = \|\cdot\|_2$, $\alpha = 1$. The hinge is 1-Lipschitz in $t = y\langle w, x \rangle$ and $|y\langle w - w', x \rangle| \leq R\|w - w'\|_2$, so $G = R$. For $\|u\|_2 \leq B$ we have $\Psi(u) = \frac{1}{2}\|u\|_2^2 \leq \frac{1}{2}B^2$.

Specializing B.3 and minimizing $\frac{\lambda}{2}B^2 + \frac{2R^2}{\lambda n}$ over λ gives $\lambda = \frac{2R}{B\sqrt{n}}$ with balanced value $\frac{2RB}{\sqrt{n}}$, and the optimization term becomes $R\sqrt{2\eta/\lambda} = \sqrt{RB\eta}n^{1/4}$.

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(u) + \frac{2RB}{\sqrt{n}} + n^{1/4}\sqrt{RB\eta}.$$

Optimization-error condition. $n^{1/4}\sqrt{RB\eta} \leq \frac{RB}{\sqrt{n}} \iff \sqrt{n}RB\eta \leq \frac{R^2B^2}{n} \iff \eta \leq \frac{RB}{n^{3/2}}$. Under $\eta \leq RB/n^{3/2}$ the excess risk is at most $\frac{3RB}{\sqrt{n}}$ — again the optimization term is a lower-order $O(n^{-3/2})$ requirement and the statistical rate $RB/\sqrt{n}$ is preserved.

Part C — A Weighted Euclidean Geometry

Linear prediction with scalar loss convex and g -Lipschitz in the prediction. $\mathcal{C}_b = \{w : |w_j| \leq b_j\}$, $|\varphi_j(x)| \leq r_j$, $Q = \text{diag}(q_1, \dots, q_d)$ with $q_j > 0$, $\|w\|_Q = \sqrt{w^\top Q w}$, $\Psi_Q(w) = \frac{1}{2}w^\top Q w$ (1-strongly convex w.r.t. $\|\cdot\|_Q$).

C.1 Geometry quantities (5 pts)

Dual norm. $\|v\|_{Q,*} = \sup_{\|w\|_Q \leq 1} \langle v, w \rangle$. Substituting $u = Q^{1/2}w$ (so $\|w\|_Q = \|u\|_2$, $w = Q^{-1/2}u$),

$$\|v\|_{Q,*} = \sup_{\|u\|_2 \leq 1} \langle Q^{-1/2}v, u \rangle = \|Q^{-1/2}v\|_2 = \sqrt{v^\top Q^{-1}v} = \sqrt{\sum_j v_j^2/q_j}.$$

Hence

$$\|\varphi(x)\|_{Q,*}^2 = \sum_{j=1}^d \frac{\varphi_j(x)^2}{q_j} \leq \sum_{j=1}^d \frac{r_j^2}{q_j}.$$

Domain radius. $\sup_{w \in \mathcal{C}_b} \Psi_Q(w) = \sup_{|w_j| \leq b_j} \frac{1}{2} \sum_j q_j w_j^2 = \frac{1}{2} \sum_{j=1}^d q_j b_j^2$, each term maximized at $w_j = \pm b_j$.

□

C.2 Fixed Q (5 pts)

As a function of w , the loss is G -Lipschitz w.r.t. $\|\cdot\|_Q$ with

$$|\ell(w) - \ell(w')| \leq g |\langle w - w', \varphi(x) \rangle| \leq g \|w - w'\|_Q \|\varphi(x)\|_{Q,*} \leq \underbrace{g \sqrt{\sum_j r_j^2/q_j}}_{=:G} \|w - w'\|_Q,$$

by Cauchy-Schwarz in the $\|\cdot\|_Q/\|\cdot\|_{Q,*}$ duality. The Week-8 theorem ($\alpha = 1$) with comparators restricted to $\mathcal{C}_b$ (so $\Psi_Q(w) \leq P := \frac{1}{2} \sum_j q_j b_j^2$) gives

$$\mathbb{E} L_{\mathcal{D}}(A_\lambda(S)) \leq \inf_{w \in \mathcal{C}_b} L_{\mathcal{D}}(w) + \lambda P + \frac{2G^2}{\lambda n}.$$

Minimizing over λ (AM–GM): $\min_{\lambda}(\lambda P + \frac{2G^2}{\lambda n}) = 2\sqrt{2G^2 P/n} = 2G\sqrt{2P/n}$. With $G^2 = g^2 \sum_j r_j^2/q_j$ and $2P = \sum_j q_j b_j^2$,

$$2G\sqrt{\frac{2P}{n}} = \frac{2g}{\sqrt{n}} \sqrt{\left(\sum_j r_j^2/q_j\right) \left(\sum_j q_j b_j^2\right)}.$$

Therefore

$$\mathbb{E} L_{\mathcal{D}}(A_{\lambda}(S)) \leq \inf_{w \in \mathcal{C}_b} L_{\mathcal{D}}(w) + \frac{2g}{\sqrt{n}} \sqrt{\left(\sum_{j=1}^d q_j b_j^2\right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j}\right)}. \quad \square$$

C.3 Best diagonal geometry (5 pts)

Minimize $\Phi(q) = (\sum_j q_j b_j^2)(\sum_j r_j^2/q_j)$ over $q_j > 0$. By Cauchy–Schwarz,

$$\left(\sum_j q_j b_j^2\right) \left(\sum_j \frac{r_j^2}{q_j}\right) \geq \left(\sum_j \sqrt{q_j b_j^2} \cdot \sqrt{\frac{r_j^2}{q_j}}\right)^2 = \left(\sum_{j=1}^d b_j r_j\right)^2,$$

with equality iff $q_j b_j^2 \propto r_j^2/q_j$, i.e. $q_j \propto r_j/b_j$. (Check $q_j = r_j/b_j$: both factors equal $\sum_j b_j r_j$.) Hence $\min_q \sqrt{\Phi(q)} = \sum_j b_j r_j$, and the best excess term is

$$\frac{2g}{\sqrt{n}} \sum_{j=1}^d b_j r_j.$$

The optimal geometry rescales each coordinate by $q_j \propto r_j/b_j$, replacing the product of ℓ_2 -type norms by the dimension-adapted ℓ_1 -type sum $\sum_j b_j r_j$. $\square$

Part D — AI Usage Report

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D.1 Where I used AI

This assignment is entirely theoretical (Parts A–C are proofs; there is no experiment or code component), so AI played the role of a *proof collaborator and LaTeX/exposition assistant*, not a code debugger. Concretely I used AI to:

- **Stress-test the proof skeleton** I had drafted on paper for A.1–A.4 (the validation oracle inequality and its adaptive-tuning consequence), checking that each “ $\leq$ ” was justified and that I had not silently swapped n_T and n_V .
- **Check algebra** in the optimizations: the AM–GM balancing of $\lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T}$ (A.3), the $c + 1/c$ discretization constant, the $n_T = n_V = n/2$ simplification (A.4), the λ choice in B.4/B.5, and the Cauchy–Schwarz step in C.3.
- **Improve exposition**, turning my margin notes into clean markdown with consistent notation.

D.2 Suggestions I accepted

1. **The “expectation of the max” route for A.1.** I initially planned a high-probability bound (union bound + Hoeffding tail, then integrate). AI suggested going directly through

the subgaussian maximal inequality $\mathbb{E} \max \leq \sigma \sqrt{2 \log(2K)}$ with $\sigma^2 = 1/(4n_V)$, which lands *exactly* on the target constant $2\sqrt{\log(2K)/(2n_V)}$ in expectation without a failure-probability term. I accepted it because it is both shorter and matches the stated bound’s constant exactly; I verified $\sigma^2 = 1/(4n_V)$ from Hoeffding’s lemma myself.

2. **Decoupling B.3 from the stability bound B.2.** My first attempt at B.3 bounded the generalization gap of $\tilde{w}_S$ using the B.2 stability constant directly, which produced an *extra* $2G\sqrt{2\eta/(\lambda\alpha)}$ term — too large to match the claim. AI pointed out the cleaner decomposition: bound $L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(w_S) + G\|\tilde{w}_S - w_S\|$ using Lipschitzness of the *true* risk, then apply the exact-RERM bound to w_S . This gives exactly the single $G\sqrt{2\eta/(\lambda\alpha)}$ term. I accepted after re-deriving it.

D.3 Suggestions I rejected or substantially modified

1. **Wrong strong-convexity constant.** AI at one point asserted that the regularized objective $F_S = L_S + \lambda\Psi$ is “ α -strongly convex.” It is $\lambda\alpha$ -strongly convex (the λ multiplies Ψ). This matters in B.1: the correct bound is $\|\tilde{w}_S - w_S\| \leq \sqrt{2\eta/(\lambda\alpha)}$, and dropping the λ would have given the wrong dependence. I corrected it.
2. **Off-by-constant in B.5.** A draft treated the constraint $\|u\|_2 \leq B$ as $\Psi(u) \leq B^2$, but $\Psi(u) = \frac{1}{2}\|u\|_2^2 \leq \frac{1}{2}B^2$. I redid the λ -optimization with the correct $\frac{1}{2}B^2$, yielding $\lambda = \frac{2R}{B\sqrt{n}}$ and the leading term $\frac{2RB}{\sqrt{n}}$ (rather than an inflated constant).
3. **Over-claiming on the “price of tuning.”** AI initially described the adaptivity cost as “ $\sqrt{\log(B_{\max}/B_{\min})/n}$.” That is the cost if $K \propto B_{\max}/B_{\min}$, but the dyadic grid has $K = O(\log(B_{\max}/B_{\min}))$, so the validation term is the *doubly*-logarithmic $\sqrt{\log \log(B_{\max}/B_{\min})/n}$. I corrected the write-up in A.4.

D.4 How I verified correctness

Since the deliverables are proofs, verification was step-level:

- **Constant tracking.** For every boxed bound I checked that the target constant emerges, not just the rate: the $2\sqrt{\log(2K)/(2n_V)}$ in A.1, the $\frac{5}{2}\sqrt{2}$ in A.3 (from $\max_{c \in [1/2, 2]}(c + 1/c) = 5/2$), the collapse $\frac{5}{2}\sqrt{2} \cdot \sqrt{2} = 5$ in A.4, and $(2\sqrt{2} + 1)$ in B.4.
- **Limit/sanity checks.** As $\eta \rightarrow 0$, B.1–B.5 reduce to the exact-RERM statements; B.4’s condition $\eta = O(n^{-3/2})$ is below the statistical $n^{-1/2}$ scale, as expected. In C.3, the optimal $q_j \propto r_j/b_j$ I verified by plugging back in to confirm both Cauchy–Schwarz factors equal $\sum_j b_j r_j$.
- **Independence bookkeeping.** The crux of A.2 is that $h_\lambda = A_\lambda(T)$ is independent of V given T ; I made sure the conditioning order (condition on T , apply A.1 over V , then $\mathbb{E}_T$) was airtight before trusting the oracle inequality.

D.5 Updates to my AI workflow (5 most recent)

1. **Added a “constant audit” pass to my proof checklist.** After the B.3/B.5 constant slips, I now make a final pass that re-derives every boxed constant independently of the AI’s text. *Why:* AI errors in these problems are almost always constant/factor errors, not structural ones.

2. **Added a “strong-convexity parameter” reminder to CLAUDE.md’s notes section.** A standing note: regularized objectives carry the λ into the strong-convexity modulus ($\lambda\alpha$, not α). *Why:* this exact mistake recurred and is easy to miss.
3. **Standardized the train/validation split notation prompt.** I now state n_T, n_V (and which sample each quantity uses) explicitly in the prompt before asking for help. *Why:* AI mixed up n_T and n_V in an early A.2 draft.
4. **“Reduce to known limit” verification habit.** I added a step that checks every approximate/perturbed bound against its exact ($\eta \rightarrow 0$) special case. *Why:* it instantly caught the spurious extra stability term in B.3.
5. **Kept the build_pdfs.sh pandoc pipeline from HW7** unchanged (theory + ai_report $\rightarrow$ combined hw8_writeup.pdf), since it has been reliable. *Why:* a stable, scripted build removes a class of last-minute formatting errors before submission.

I used AI for proof-checking, algebra verification, and exposition only; the proof strategies and all final corrections are my own.